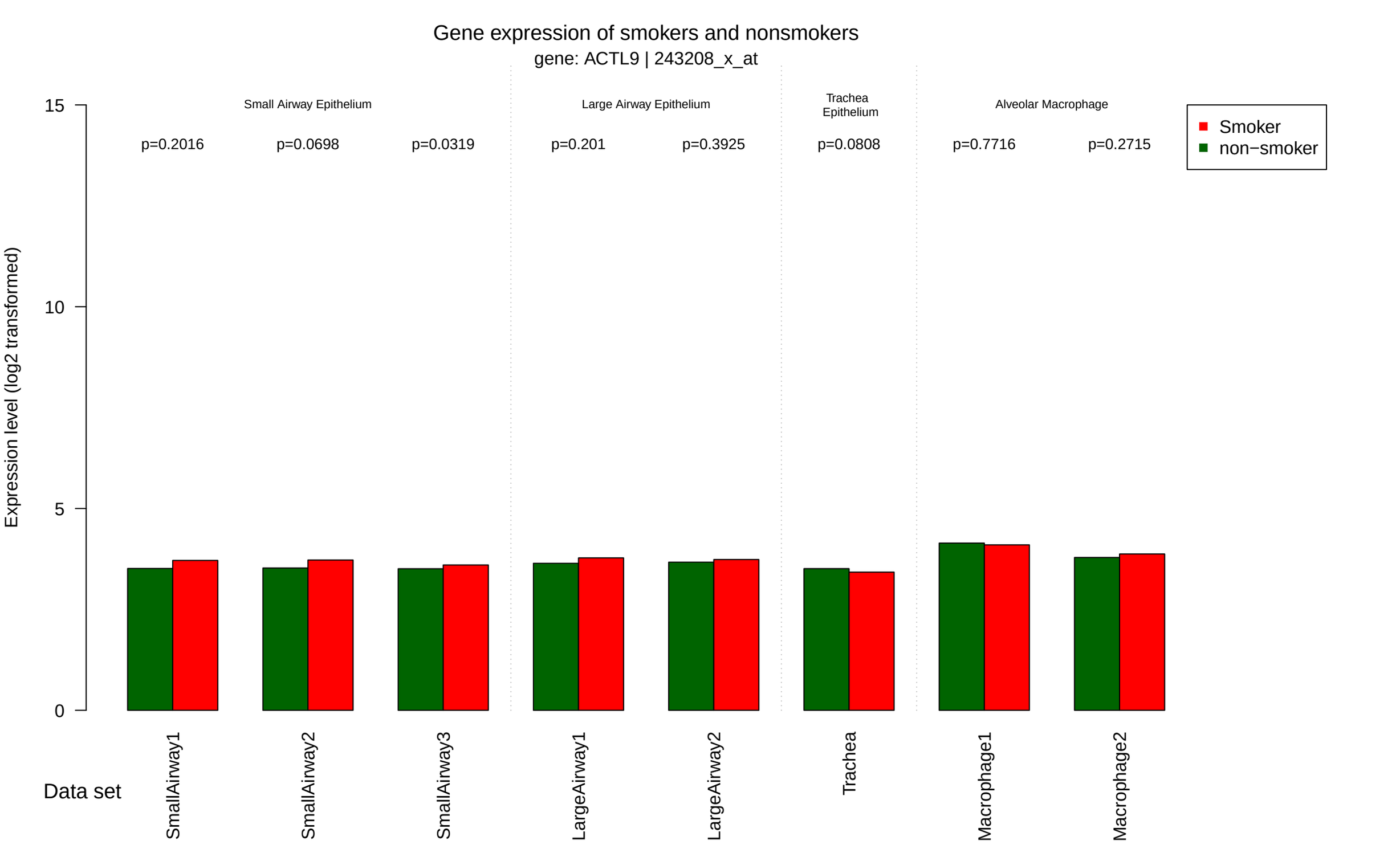
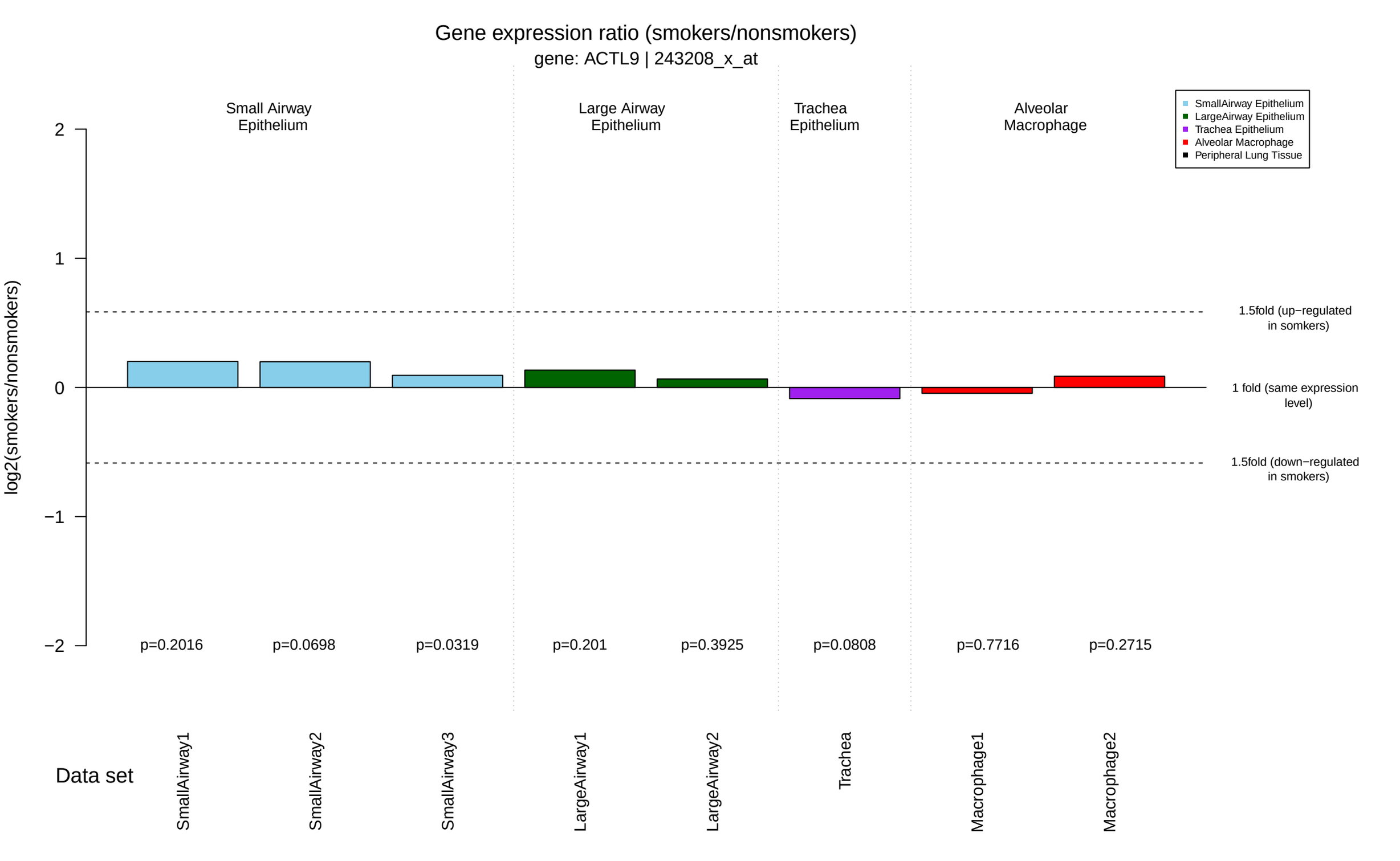
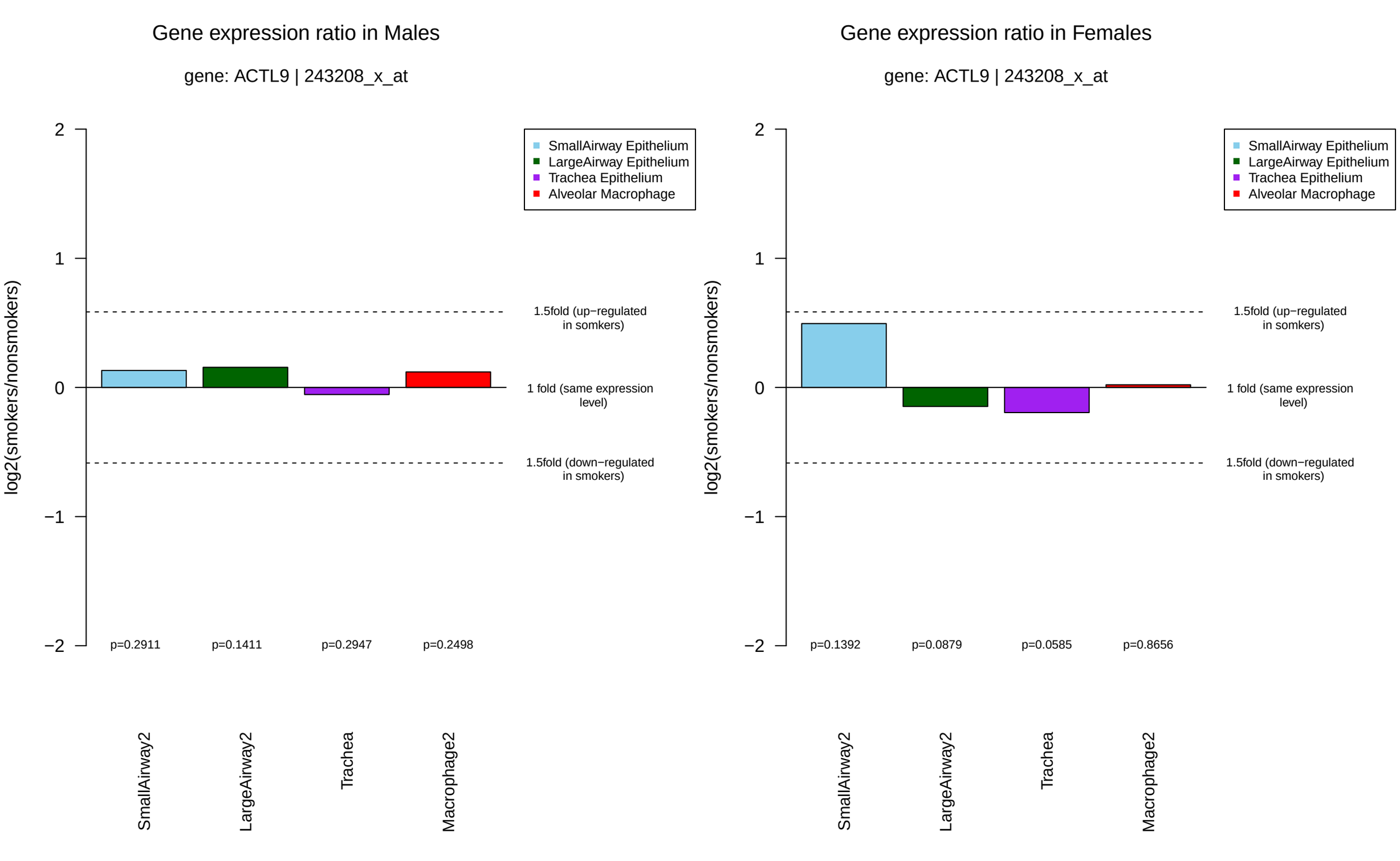
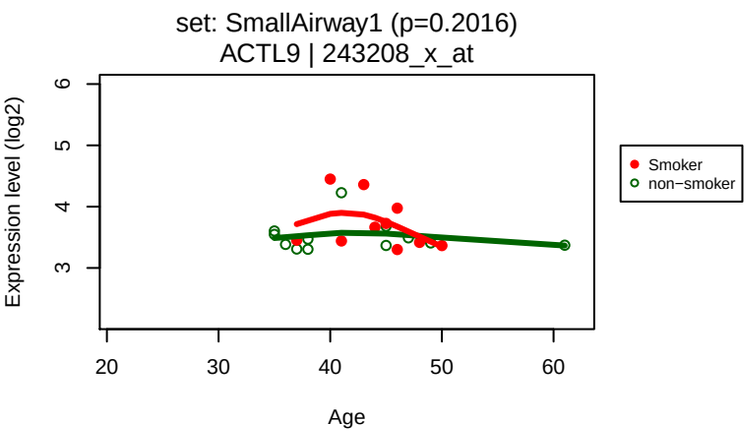
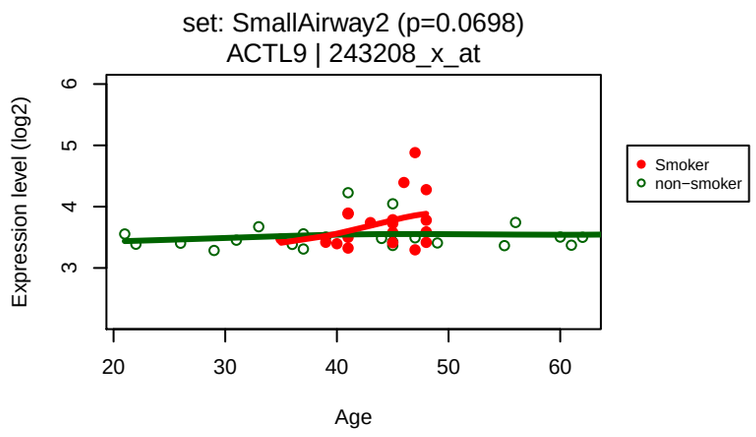
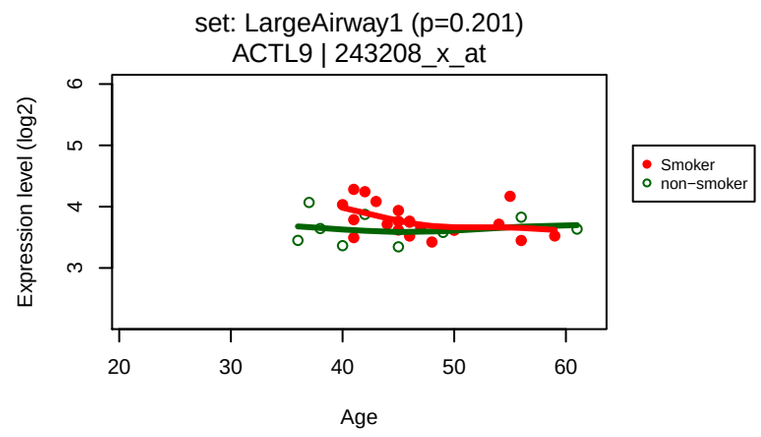

Gene expression of smokers and nonsmokers
gene: ACTL9 | 243208_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2016
p=0.0698
p=0.0319
p=0.201
p=0.3925
p=0.0808
p=0.7716
p=0.2715
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

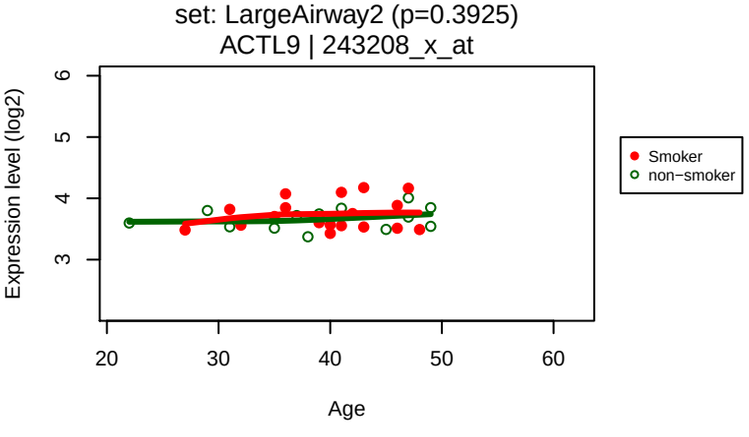

Gene expression ratio (smokers/nonsmokers)
gene: ACTL9 | 243208_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2016
p=0.0698
p=0.0319
p=0.201
p=0.3925
p=0.0808
p=0.7716
p=0.2715
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

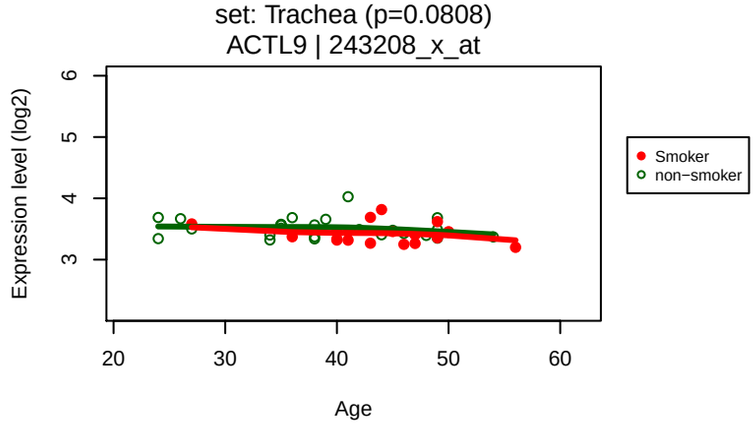

Gene expression ratio in Males
gene: ACTL9 | 243208_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2911
p=0.1411
p=0.2947
p=0.2498
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTL9 | 243208_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1392
p=0.0879
p=0.0585
p=0.8656
SmallAirway2
LargeAirway2
Trachea
Macrophage2

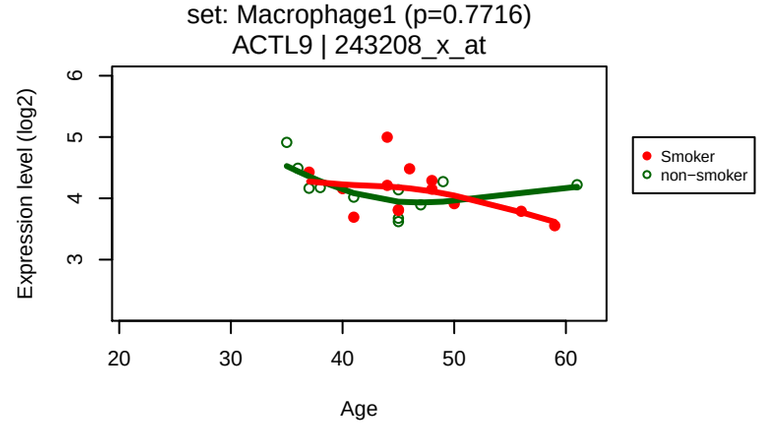

set: SmallAirway1 (p=0.2016)
ACTL9 | 243208_x_at
Expression level (log2)
Age
Smoker
non−smoker

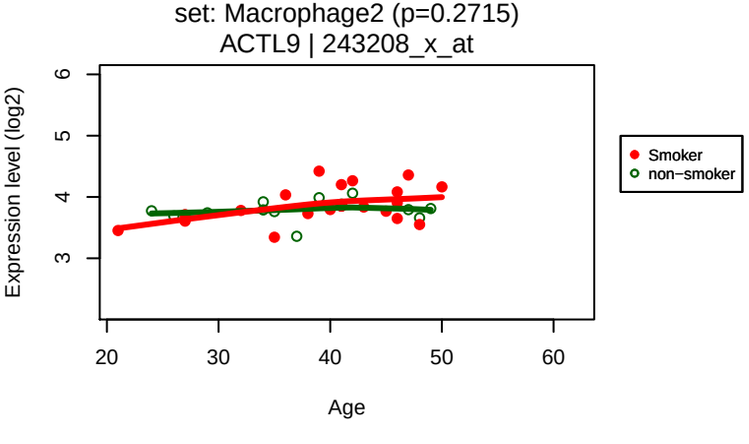

set: SmallAirway2 (p=0.0698)
ACTL9 | 243208_x_at
Expression level (log2)
Age
Smoker
non−smoker

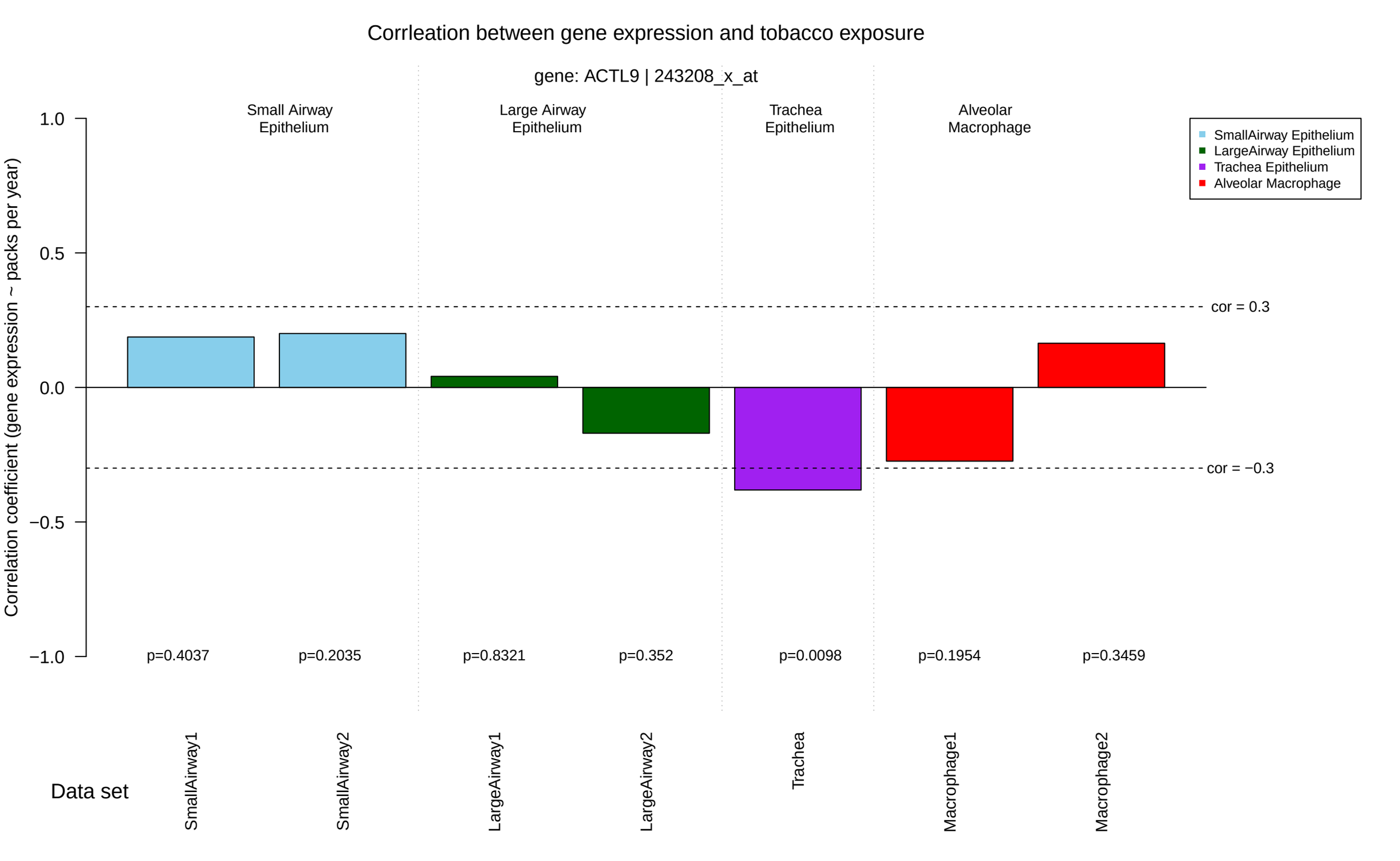

set: LargeAirway1 (p=0.201)
ACTL9 | 243208_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3925)
ACTL9 | 243208_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0808)
ACTL9 | 243208_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7716)
ACTL9 | 243208_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2715)
ACTL9 | 243208_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTL9 | 243208_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4037
p=0.2035
p=0.8321
p=0.352
p=0.0098
p=0.1954
p=0.3459
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2